

Subject Description Form

Subject Code	ITC502
Subject Title	Quality Assurance in Textiles & Clothing
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	1. To provide comprehensive study of current quality management practice in the textile and clothing industry 2. To provide the knowledge necessary for the application and implementation of quality accreditation scheme, such as ISO 9000 to a textile and clothing organisations 3. To provide quality management and appropriate continuous improvement methodology and techniques to analyse problems and generate solutions in improving the business and operational company performance in textile and clothing
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. Evaluate the status of the textile and clothing enterprises in terms of their quality management practice. 2. Understand and evaluate the mechanism and operation of international quality management systems 3. Apply quality management and appropriate continuous improvement methodology and techniques, such as six sigma, to define, measure, analyse, improve and control the operational performance of enterprises.
Subject Synopsis/ Indicative Syllabus	1. <u>Quality Management Systems in Textile and Clothing Organisations</u> Installation, implementation and monitoring quality systems. 2. <u>Overview of Total Quality Management Concept and Contemporary Management Tools</u> Evolution of Quality Models, Six Sigma, Balanced Score Card, Cost of Quality Concept, Project Identification, Project Charter Writing. 3. <u>Identification of Improvement Area and Measurement in Textile and Clothing Business</u> Voice of Customer

	4. <u>Techniques for Analyzing the Current Situation textile and clothing organisations</u> SIPOC Diagram, Process Mapping, Value-added Analysis, Root Cause Identification and Verification, Sigma Calculation, Process Capability Study, Statistical process control. 5. <u>Techniques for Improvement</u> Seven Management Tools, Quality Function Deployment, Failure Modes and Effects Analysis, Process Control Plan/ 6. <u>Mechanism of Continuous Improvement</u> Introduction of DMAIC methodology, Deployment strategies in textile and clothing organization.																																	
Teaching/Learning Methodology	This subject will be delivered based on student-centred learning method. Lectures and team-based simulation projects are designed for the students. Lectures will be supplemented by the use of case studies relevant to the subject. Students are required to form project teams to apply the continuous improvement methodology to achieve specific milestones so as to solve a problem assigned. Their deliverables will be used for coursework.																																	
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>1</th><th>2</th><th>3</th></tr><tr><td>1. Coursework</td><td></td><td></td><td></td><td></td></tr><tr><td> a) Individual assignments</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td> b) Team-based project</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="3"></td></tr></table> As the coursework and final examination questions are designed to be more application oriented with the consideration of both theories and real-life scenarios, the three learning outcomes can be assessed. Students will be required to deliver an individual project to apply appropriate quality management for identifying, formulating and implementing business strategies so as to improve business performance in their workplaces based on the concepts learned in the quality assurance and total quality management. Students are also required to participate a team-based project in which a systematic approach with appropriate quality, operational and management techniques should be applied for analyzing the problem assigned and generating solutions for improvement.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			1	2	3	1. Coursework					a) Individual assignments	25%	✓	✓	✓	b) Team-based project	25%	✓	✓	✓	2. Examination	50%	✓	✓	✓	Total	100%			
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2. Examination	50%	✓	✓	✓																														
Total	100%																																	

Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Seminars/Tutorials	9 Hrs.
	Other student study effort:	
	▪ Student Self-Study	53 Hrs.
	▪ Coursework	43 Hrs.
	Total student study effort	135 Hrs.
Reading List and References	<u>Books</u> Foster, Thomas S (2017) <i>Managing Quality: Integrating the Supply Chain</i> 6th edition, Pearson Education International, Sixth Edition, Pearson, Boston Oakland, J. S. (2014), <i>Total Quality Management and Operational Excellence: Text with Cases</i>. 4th Edition, Routledge, Oxon. <u>References</u> De Feo, Joseph A. and Gryna, Frank M. (2015) <i>Juran's Quality Management and Analysis</i>, 6th Edition, McGraw Hill, New York. Dentch, M.P. (2017) <i>The ISO 9001:2015 Implementation Handbook: Using the Process Approach to Build a Quality Management System</i>, ASQ Quality Press, Wisconsin. Evans, J.R. and Lindsay, W.M. (2015) <i>An Introduction to Six Sigma & Process Improvement</i>, 2nd Edition, Cengage, Stamford. Evans, J.R. (2020), <i>Managing for quality and performance excellence</i>, 10th Edition, Cengage Learning, Boston, MA. Goetsch, D.L. & Davis, S. B. (2016), <i>Quality Management: Introduction to Total Quality Management for Production, Processing and Services</i>, 8th Edition, Prentice Hall, Upper Saddle River, NJ. Hoyle, D. (2018) <i>ISO 9000 quality systems handbook: increasing the quality of an organization's outputs</i>, 7th Edition, New York, Routledge. New York. Juran, J.M. (ed.) (1995), <i>A history of managing for quality: the evolution, trends, and future directions of managing for quality</i>, ASQ Quality Press, Wisconsin. Merrill, P. (2009) <i>Do it Right the Second Time: Benchmarking Best Practices in the Quality Change Process</i>, 2nd Edition, ASQ Quality Press, Wisconsin. Nayatani Y. (1994), <i>The Seven New QC Tools, Practical Applications for Managers</i>, 3A Corporation, Tokyo. <u>Periodicals</u> International Journal of Quality and Reliability Management Quality Progress The Journal for Quality and Participation Total Quality Management Total Quality Management & Business Excellence	